

Student Number: _____ Class Teacher: _____

St George Girls High School

Trial Higher School Certificate Examination

2018



Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using black pen.
- Board-approved calculators may be used.
- A reference sheet is provided.
- In Questions 11 – 15, show relevant mathematical reasoning and/or calculations.
- Marks may not be awarded for incomplete or poorly presented solutions.

Section I	/10
Section II	
Question 11	/12
Question 12	/12
Question 13	/12
Question 14	/12
Question 15	/12
Total	/70

Total Marks – 70

Section 1

Pages 3 – 6

10 marks

- Attempt Questions 1 – 10.
- Allow about 15 minutes for this section.
- Answer on the multiple choice answer sheet provided at the back of this paper.

Section II

Pages 7 – 13

60 marks

- Attempt Questions 11 – 15.
- Allow about 1 hour and 45 minutes for this section.
- Begin each question in a new writing booklet.

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

- 1 What is the size of the acute angle between $y = 3x$ and $y = -2x + 1$, correct to the nearest degree?

- (A) 36°
- (B) 135°
- (C) 11°
- (D) 45°

- 2 Simplify $\frac{\cos 3\alpha}{\sin \alpha} + \frac{\sin 3\alpha}{\cos \alpha}$.

- (A) $\tan 2\alpha$
- (B) $\cot 2\alpha$
- (C) $2 \tan 2\alpha$
- (D) $2 \cot 2\alpha$

- 3 What is the maximum value of the function $y = \sin x - \sqrt{3} \cos x + 1$.

- (A) 1
- (B) 2
- (C) 3
- (D) 4

- 4 The area of a rectangular bar is $A = 50L - L^2$, where L is the length of the bar. The bar is heated and its length increases at the rate of 0.08 cm/min. At what rate is the area of this bar increasing when $L = 15$ cm?

- (A) 3.6 cm²/min
- (B) 2.4 cm²/min
- (C) 1.6 cm²/min
- (D) 0.8 cm²/min

- 5 Find the value of a , if $(x + a)$ is a factor of $P(x) = x^3 + ax^2 + 2x + 1$.

- (A) $-\frac{1}{2}$
- (B) ± 1
- (C) $\frac{1}{2}$
- (D) $\pm \frac{1}{\sqrt{2}}$

- 6 If $y = \pi^x$, find $\frac{dy}{dx}$ when $x = 1$.

- (A) 1
- (B) π
- (C) $\pi \ln \pi$
- (D) $x\pi - \ln x$

7 What is the exact value of $\int_0^4 \frac{dx}{x^2 + 16}$?

(A) $-\frac{\pi}{4}$

(B) $\frac{\pi}{16}$

(C) $\frac{\pi}{8}$

(D) $\frac{\pi}{4}$

8 Which of the following is the range of the function $y = 2 \sin^{-1} x + \frac{\pi}{2}$?

(A) $y \in \mathbb{R}: -\pi \leq y \leq \pi$

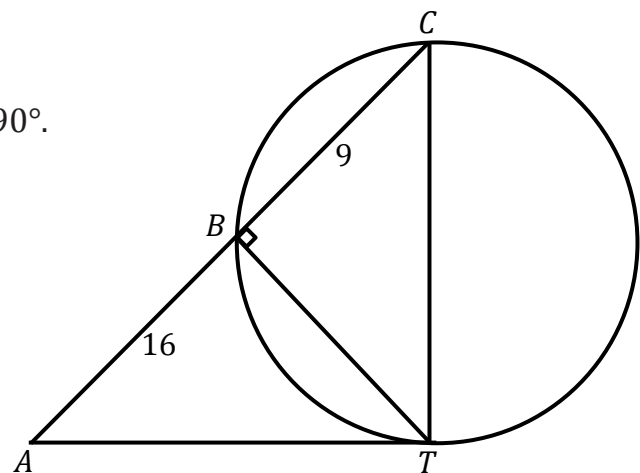
(B) $y \in \mathbb{R}: -\pi \leq y \leq \frac{3\pi}{2}$

(C) $y \in \mathbb{R}: -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

(D) $y \in \mathbb{R}: -\frac{\pi}{2} \leq y \leq \frac{3\pi}{2}$

- 9 AT is tangent to the circle at T .
 AC cuts the circle at B .
 $AB = 16$, $BC = 9$, and $\angle TBC = 90^\circ$.

Which is the length of TC , correct to the nearest whole number?



- (A) 9
(B) 15
(C) 22
(D) 25

10 Which of the following is an expression for $\tan(\cos^{-1} x)$?

(A) $\sqrt{1 - x^2}$

(B) $\frac{\sqrt{1 - x^2}}{x}$

(C) $\frac{x}{\sqrt{1 + x^2}}$

(D) $\frac{\sqrt{1 + x^2}}{x}$

End of Section I

Section II

60 marks

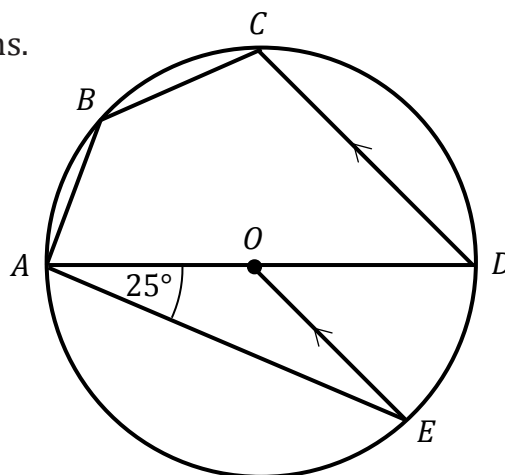
Attempt Questions 11 – 15

Allow about 1 hour and 45 minutes for this section

Answer each question in the appropriate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

- | Question 11 | (12 marks) Use a separate writing booklet | Marks |
|--------------------|---|--------------|
| (a) | Solve $ x - 7 < 3 - x $. | 2 |
| (b) | Find $\frac{d}{dx}(x \sin^{-1} x)$. | 2 |
| (c) | The point R divides the interval from $A(-10, 8)$ to $B(-5, -1)$ externally in the ratio 1:2.
Find the coordinates of R . | 2 |
| (d) | Find $\int \sec^2 x \tan x \, dx$. | 1 |
| (e) | Evaluate $\lim_{x \rightarrow 0} \frac{\sin(\pi + x)}{x}$. | 2 |
| (f) | The points A, B, C , and D lie on a circle with centre O , such that AD is a diameter. The point E lies on the circle so that OE is parallel to CD . $\angle OAE = 25^\circ$.
Find the size of $\angle ABC$, giving reasons. | 3 |



(a) Solve $\frac{4x-1}{x+2} \geq 1$. 3

$$\int_0^3 \frac{x}{\sqrt{x+1}} dx.$$

iii) Find the acceleration of the particle in terms of χ . 1

(d) Find the general solution of $2 \sin^2 x - 1 = 0$.
Give your answer in radians.

Question 13 (12 marks) Use a separate writing booklet Marks

- (a) Two points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ (where $p > q > 0$) lie on the parabola $x^2 = 4ay$.
- i) Derive the equation of the tangent to the parabola at P . 2
- ii) Find the coordinates of the point of intersection T of the tangents to the parabola at P and Q . 2
- iii) Show that if the tangents at P and Q intersect at 45° , then $p - q = 1 + pq$. 1
- iv) Find the locus of T by evaluating the expression $x^2 - 4ay$ at T and using the result in part (iii). 2
- (b) Express $2 \sin x - 3 \cos x$ in the form $R \cos(x + \alpha)$, where $R > 0$ and $0 \leq \alpha \leq 2\pi$. Give the value for α to 1 decimal place. 3
- (c) At time t years after the start of the year 2000, the number of individuals in a population is given by $N = 80 + Ae^{0.1t}$, for some constant $A > 0$. 2

If there were 100 individuals in the population at the start of the year 2000, during which year is the population expected to reach 200?

Question 14 (12 marks) Use a separate writing booklet Marks

- (a) i) Use the sum of the terms of an arithmetic series to show that 1
- $$(1 + 2 + 3 + \cdots + n)^2 = \frac{1}{4}n^2(n + 1)^2.$$
- ii) Prove the following expression by mathematical induction: 3
- $$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2 \text{ for all integers } n \geq 1.$$
- (b) i) Divide the polynomial $f(x) = 2x^4 - 10x^3 + 12x^2 + 2x - 3$ by 2
- $$g(x) = x^2 - 3x + 1.$$
- ii) Hence express $f(x)$ in the form $f(x) = g(x)q(x) + r(x)$, 1
- where $q(x)$ and $r(x)$ are polynomials, and $r(x)$ has degree less than 2.
- iii) Hence show that $f(x)$ and $g(x)$ have no zeroes in common. 1

Question 14 continues on next page

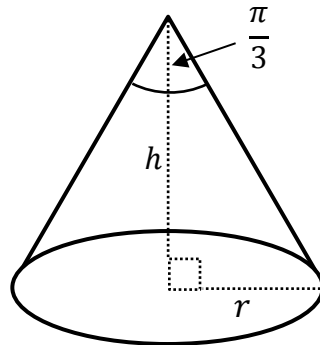
Question 14 continued

Marks

- (c) Coal is poured at a constant rate of 1.5 cubic metres per second from a ship onto a conical pile on a dock.

The angle at the apex of the cone is a constant $\frac{\pi}{3}$ radians.

At time t seconds the height of the cone is h metres and the radius of the base is r metres.



- (i) Show that $r = \frac{h}{\sqrt{3}}$. 1
- (ii) Show that the volume of the pile, $V \text{ m}^3$, is given by 1
- $$V = \frac{\pi h^3}{9}.$$
- (iii) Hence find the exact rate at which the height of the pile is increasing when the height of the pile is 6 metres. 2

Question 15 (12 marks) Use a separate writing booklet Marks

- (a) Use the substitution $t = \tan \frac{x}{2}$ to show that 2

$$\frac{\cos x}{1 - \sin x} = \tan \left(45^\circ + \frac{x}{2} \right).$$

- (b) A firework is launched from a point $(0, 0)$ with a velocity of V m/s at an angle of θ to the horizontal.

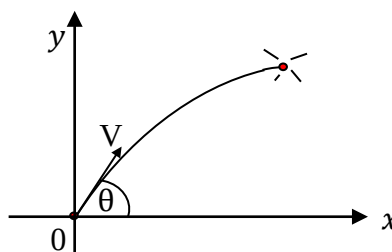
This firework explodes when it reaches its maximum height.

Use the axes as shown, assume that there is no air resistance, and that the position of the firework t seconds after being launched is given by the equations:

$$x = Vt \cos \theta \text{ and}$$

$$y = Vt \sin \theta - \frac{1}{2}gt^2$$

(Do NOT prove these results)



- i) Show that the maximum height reached where the firework explodes is given by 2
- $$y = \frac{V^2 \sin^2 \theta}{2g}.$$

- ii) A second firework is launched from a point $(0,0)$ with a velocity of $\frac{7V}{10}$ m/s at an angle of 2θ to the horizontal and also explodes when it reaches its maximum height.
 Given that the two fireworks reach the same maximum height:

- α) Show that $\cos \theta = \frac{5}{7}$. 3

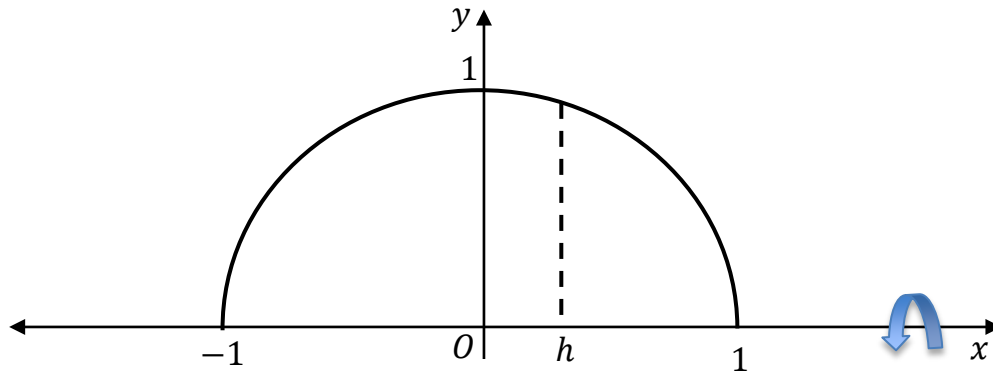
- β) If $v^2 = 98g$, find the maximum height reached by the two fireworks. 1

Question 15 continues on next page

Question 15 continued

Marks

- (c) The region enclosed by the semicircle $y = \sqrt{1 - x^2}$ and the x -axis is to be divided into two pieces by the line $x = h$, where $0 \leq h < 1$.



The two pieces are rotated about the x -axis to form solids of revolution. The value of h is chosen so that the volume of the solids are in the ratio 3:2.

- i) Given that the volumes are in the ratio 3:2, show that $5h^3 - 15h + 2 = 0$. 3
- ii) Given $h_1 = 0$ as the first approximation for h , use one application of Newton's Method to find a second approximation for h . 1

End of examination

MATHEMATICS EXTENSION 1 – MULTIPLE CHOICE

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
① $y = 3x \Rightarrow m_1 = 3$ $y = -2x + 1 \Rightarrow m_2 = -2$ $\tan \theta = \left \frac{3 - (-2)}{1 + 3(-2)} \right $ $= \left \frac{5}{-5} \right $ $= 1$ $\therefore \theta = 45^\circ$ (D)		
② $\frac{\cos 3\alpha}{\sin \alpha} + \frac{\sin 3\alpha}{\cos \alpha} = \frac{\cos 3\alpha \cos \alpha + \sin 3\alpha \sin \alpha}{\sin \alpha \cos \alpha}$ $= \frac{\cos(3\alpha - \alpha)}{\frac{1}{2} \sin 2\alpha}$ $= \frac{2 \cos 2\alpha}{\sin 2\alpha}$ $= 2 \cot 2\alpha$ (D)		
③ $\sin x + \sqrt{3} \cos x$ can be expressed in the form $R \sin(x + \alpha)$, where $R = \sqrt{1^2 + (\sqrt{3})^2} = 2$ $\therefore y = 2 \sin(x + \alpha) + 1$ $\therefore$ maximum value is 3 (C)		
④ $A = 50L - L^2$ $\frac{dA}{dL} = 50 - 2L \quad \text{when } L = 15, \frac{dA}{dL} = 20$ Also $\frac{dL}{dE} = 0.08$ $\frac{dA}{dE} = \frac{dA}{dL} \times \frac{dL}{dE}$ $= 20 \times 0.08$ $= 1.6 \text{ cm}^2/\text{min}$ (C)		

MATHEMATICS EXTENSION I – MULTIPLE CHOICE

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
⑤ If $(x+a)$ is a factor, $p(-a)=0$ i.e. $-a^3 + a^3 - 2a + 1 = 0$ $2a = 1$ $a = \frac{1}{2}$ (C)		
⑥ $y = \pi^x$ $= (e^{\ln \pi})^x$ $= e^{\ln \pi x}$ $\frac{dy}{dx} = \ln \pi \times e^{\ln \pi x}$ $= \ln \pi \times \pi^x$ when $x=1$, $\frac{dy}{dx} = \ln \pi \times \pi$ $= \pi \ln \pi$ (C)		
⑦ $\int_0^4 \frac{dx}{x^2+16} = \int_0^4 \frac{dx}{x^2+4^2}$ $= \left[\frac{1}{4} \tan^{-1} \frac{x}{4} \right]_0^4$ $= \frac{1}{4} (\tan^{-1} 1 - \tan^{-1} 0)$ $= \frac{1}{4} \left(\frac{\pi}{4} - 0 \right)$ $= \frac{\pi}{16}$ (B)		
⑧ $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$ $-\pi \leq 2 \sin^{-1} x \leq \pi$ $-\frac{\pi}{2} \leq 2 \sin^{-1} x + \frac{\pi}{2} \leq \frac{3\pi}{2}$ (D)		

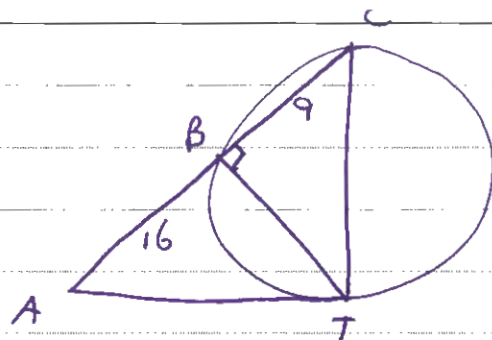
MATHEMATICS EXTENSION I – MULTIPLE CHOICE

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

9



$$AT^2 = 16 \times 25 \quad (\text{square of tangent equals product of intercepts on secant})$$

$$= 400$$

$$AT = 20$$

~~Q14~~ CT is a diameter ($\angle TBC = 90^\circ$)
 $\therefore CT \perp AT$ (radius perpendicular to tangent)

$$\therefore AC^2 = TC^2 + AT^2 \quad (\text{Pythagoras' theorem})$$

$$TC^2 = 25^2 - 20^2$$

$$= 225$$

$$TC = 15$$

(B)

10 let $\theta = \cos^{-1}x$

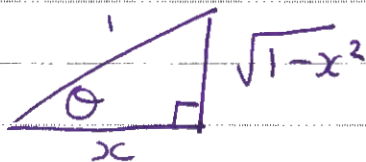
$$\text{ie } \cos \theta = x$$

$$\therefore \sin \theta = \sqrt{1-x^2}$$

$$\therefore \tan \theta = \frac{\sqrt{1-x^2}}{x}$$

$$\therefore \tan(\cos^{-1}x) = \frac{\sqrt{1-x^2}}{x}$$

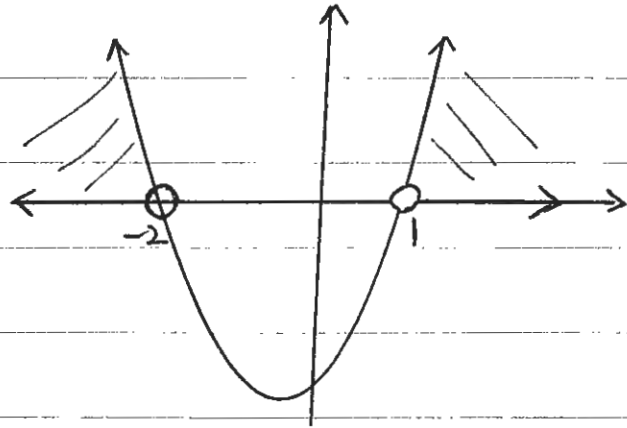
(B)



MATHEMATICS EXTENSION I – QUESTION 11

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) $x-7 < 3-x$ i.e. distance from 7 < distance from 3  $\therefore x > 5$	2	Finding $x=5$ as the critical point earned 1 mark.
b) let $u=x$ $v=\sin^{-1}x$ $u'=1$ $v'=\frac{1}{\sqrt{1-x^2}}$	1	
$\frac{d}{dx}(x \sin^{-1}x) = \sin^{-1}x + \frac{x}{\sqrt{1-x^2}}$	1	
c) $R = \left(\frac{2(-10) - 1(-5)}{2-1}, \frac{2(8) - 1(-1)}{2-1} \right)$ $= (-15, 17)$	1	Correct substitution into correct formula
d) $\int \sec^2 x \tan x \, dx = \frac{1}{2} \tan^2 x + C$	1	
e) $\lim_{x \rightarrow 0} \frac{\sin(\pi+x)}{x} = \lim_{x \rightarrow 0} \frac{-\sin x}{x} \quad \left(\begin{array}{l} \text{since } \sin(\pi+x) \\ = -\sin x \end{array} \right)$ $= -1$	1	or $\sin \pi \frac{\cos x}{x} + \cos \pi \frac{\sin x}{x}$
f) $\angle EOD = 2 \times \angle EAO$ (angle at centre twice angle at circumference) $= 50^\circ$ $\angle CDO = \angle EOD$ (alternate angles on parallel lines) $= 50^\circ$ $\angle ABC + \angle CDO = 180^\circ$ (opposite angles of cyclic quadrilateral are supplementary) $\angle ABC = 180^\circ - 50^\circ$ $= 130^\circ$	1	NB: you cannot assume B lies on EO produced.

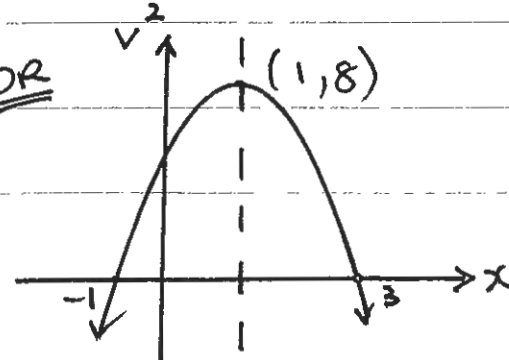
MATHEMATICS EXTENSION 1 TRIAL HSC 2018 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(a) $x \neq -2$ $(x+2)^x \times \frac{4x-1}{x+2} \geq 1 \times (x+2)^2$ $(x+2)(4x-1) \geq (x+2)^2$ $(x+2)(4x-1) - (x+2)^2 \geq 0$ $x+2 [4x-1 - (x+2)] \geq 0$ $x+2 [4x-1 - x-2] \geq 0$ $(x+2)(3x-3) \geq 0$ $3(x+2)(x-1) \geq 0$  $\therefore \underbrace{x < -2}_{\textcircled{1}} \text{ OR } \underbrace{x \geq 1}_{\textcircled{1}}$		• this question was mostly well done. • be careful with algebra! • make the statement about the denominator $x \neq -2$!!! - Hence $x < -2$ is the solution not $x \leq -2$. ① for progress to answer • be careful how you draw your parabola ie when its $\cup$ OR $\cap$ depending on your working.
③ provides correct solution with working		
② correctly provides $x \leq -2$ OR $x \geq 1$ with working		
① demonstrates some progress		

MATHEMATICS EXTENSION 1 TRIAL HSC 2018 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(b) $u = x+1 \Rightarrow x = u-1$ $\frac{du}{dx} = 1$ when $x=0, u=1$ $\therefore du = dx$ $x=3, u=4$ (1) for these steps		
$\int_0^3 \frac{x}{\sqrt{x+1}} dx$		
$= \int_1^4 \frac{u-1}{\sqrt{u}} du$		
$= \int_1^4 \frac{u}{u^{\frac{1}{2}}} - \frac{1}{u^{\frac{1}{2}}} du$		
$= \int_1^4 u^{\frac{1}{2}} - u^{-\frac{1}{2}} du$		(1) for these lines or equivalent
$= \left[\frac{2}{3} u^{\frac{3}{2}} - 2u^{\frac{1}{2}} \right]_1^4$		(1) for integrating correctly
$= \frac{2}{3}(4)^{\frac{3}{2}} - 2(4)^{\frac{1}{2}} - \left(\frac{2}{3} - 2 \right)$		
$= \frac{16}{3} - 4 - \left(\frac{2}{3} - 2 \right)$		
$= \frac{8}{3} - (1) \text{ for correct answer}$		
• this question was well done!		

MATHEMATICS EXTENSION 1 TRIAL HSC 2018 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(c) $v^2 = 6 + 4x - 2x^2$ (i) two extremes occur when $v=0$: $-2x^2 + 4x + 6 = 0$ $-2(x^2 - 2x - 3) = 0$ $(x-3)(x+1) = 0$ $\therefore x = 3 \text{ or } x = -1$ $\therefore$ particle oscillates between -1 and 3.		
		① provides both correct answers
(ii) maximum speed occurs at the centre of motion ie when $x=1$ $v^2 = 6 + 4(1) - 2(1)^2$ $= 6 + 4 - 2$ $= 8$ max. speed $= v = \sqrt{8}$ $= 2\sqrt{2} \text{ m/s}$		① provides correct answer
<u>OR</u>  $v^2 = 8$ $\therefore$ max. speed $= v$ $= \sqrt{8}$ $= 2\sqrt{2} \text{ m/s}$		

MATHEMATICS EXTENSION 1 TRIAL HSC 2018 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(iii) $a = \ddot{x} = \frac{d}{dx} \left(\frac{1}{2}v^2 \right)$ $= \frac{d}{dx} (3 + 2x - x^2)$ $\therefore a = 2 - 2x$ — ① provides correct answer • this question was well done • watch calculations • no half marks!		
(d) $2\sin^2 x - 1 = 0$ $2\sin^2 x = 1$ $\sin^2 x = \frac{1}{2}$ $\sin x = \pm \frac{1}{\sqrt{2}}$ For $\sin x = \frac{1}{\sqrt{2}}$: $x = n\pi + (-1)^n \sin^{-1} \left(\frac{1}{\sqrt{2}} \right)$ $x = n\pi + (-1)^n \frac{\pi}{4} \quad \text{where } n \text{ is an integer.}$		• mostly well done! • use reference sheet • don't forget where $n \in \mathbb{Z}$ or where n is an integer. • no $\frac{1}{2}$ marks.
For $\sin x = -\frac{1}{\sqrt{2}}$: $x = n\pi + (-1)^n \sin^{-1} \left(-\frac{1}{\sqrt{2}} \right)$ $x = n\pi + (-1)^n \left(-\frac{\pi}{4} \right) \quad \text{where } n \text{ is an integer.}$ OR $x = n\pi - (-1)^n \frac{\pi}{4}$		① mark ① mark

MATHEMATICS EXTENSION 1 TRIAL HSC 2018 – QUESTION 12

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS				
<u>OR</u> (d) $2\sin^2 x - 1 = 0$ $2\sin^2 x = 1$ $\sin^2 x = \frac{1}{2}$ $\sin x = \pm \frac{1}{\sqrt{2}}$ (ie all 4 quadrants) $\therefore x = n\pi \pm \frac{\pi}{4}$ where $n \in \mathbb{Z} \rightarrow \textcircled{1}$	<table><tr><td>-225° 135°</td><td>-315° 45°</td></tr><tr><td>225° -135°</td><td>315° -45°</td></tr></table>	-225° 135°	-315° 45°	225° -135°	315° -45°	
-225° 135°	-315° 45°					
225° -135°	315° -45°					
<u>OR</u> $2\sin^2 x - 1 = 0$ $-\cos 2x = 0$ $\cos 2x = 0$ $2x = 2n\pi \pm \cos^{-1} 0$ $2x = 2n\pi \pm \frac{\pi}{2} \rightarrow \textcircled{1}$ mark $x = \frac{1}{2} \left(2n\pi \pm \frac{\pi}{2} \right)$ $\therefore x = n\pi \pm \frac{\pi}{4}$ where $n \in \mathbb{Z}$ $\rightarrow \textcircled{1}$ mark	$\bullet \cos 2x = 1 - 2\sin^2 x$					
NOTE: $\frac{\frac{\pi}{2}}{2} = \frac{\pi}{4}$ $\nmid \frac{1}{2} \times \frac{\pi}{2} = \frac{\pi}{4}$						

MATHEMATICS EXTENSION 1 TRIAL HSC 2018 – QUESTION 12

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

OR The really long way !!!

$$2\sin^2 x - 1 = 0$$

$$2\sin^2 x = 1$$

$$\sin^2 x = \frac{1}{2}$$

$$\sin x = \pm \frac{1}{\sqrt{2}}$$

For $\sin x = +\frac{1}{\sqrt{2}}$:

$$x = \left(\sin^{-1} \frac{1}{\sqrt{2}}\right) + 2n\pi \quad \text{OR} \quad x = \left(\pi - \sin^{-1} \frac{1}{\sqrt{2}}\right) + 2n\pi$$

$$x = \frac{\pi}{4} + 2n\pi$$

$$x = \left(\pi - \frac{\pi}{4}\right) + 2n\pi$$

$$x = \frac{3\pi}{4} + 2n\pi$$

①
mark

For $\sin x = -\frac{1}{\sqrt{2}}$:

$$x = \sin^{-1} \left(-\frac{1}{\sqrt{2}}\right) + 2n\pi \quad \text{OR} \quad x = \pi - \sin^{-1} \left(-\frac{1}{\sqrt{2}}\right) + 2n\pi$$

$$x = -\frac{\pi}{4} + 2n\pi$$

$$= \pi - \left(-\frac{\pi}{4}\right) + 2n\pi$$

$$x = \frac{5\pi}{4} + 2n\pi$$

①
mark

where n is an integer!

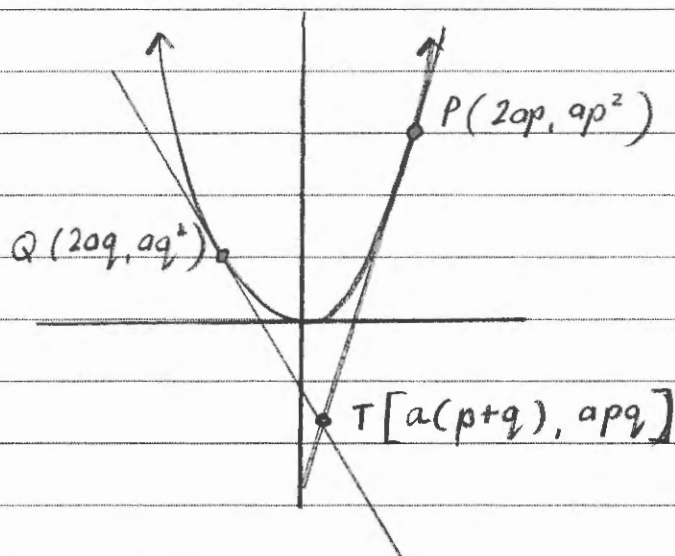
MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

a)



$$y = \frac{x^2}{4a}$$

$$y' = \frac{x}{2a} \quad x = 2ap$$

$$m = \frac{2ap}{2a}$$

$$m = p$$

}

— ①

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2 \quad \dots \textcircled{1}$$

}

①

(ii) Similarly $y = qx - aq^2 \quad \dots \textcircled{2}$

Solve ① and ② simultaneously

$$px - ap^2 = qx - aq^2$$

$$px - qx = ap^2 - aq^2$$

$$(p - q)x = a(p^2 - q^2)$$

$$x = \frac{a(p - q)(p + q)}{(p - q)}$$

$$x = a(p + q)$$

— ①

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$y = px - ap^2$$

$$y = p \times a(p+q) - ap^2$$

$$= ap^2 + apq - ap^2$$

$$y = apq$$

– ①

$$\therefore T[a(p+q), apq]$$

(iii)

$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right| \quad m_1 = p \quad m_2 = q$$

$$\tan 45^\circ = \left| \frac{p - q}{1 + pq} \right|$$

They needed to have $\tan 45^\circ$ not just "1"

$$\left| \frac{p - q}{1 + pq} \right| = 1$$

Now $p > q > 0$
 $\therefore p - q > 0$
 and $1 + pq > 0$

$$\frac{p - q}{1 + pq} = 1$$

$$p - q = 1 + pq$$

①

please stress that removing the absolute value signs should be explained.

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

(iv) $x^2 - 4ay$ $T[a(p+q), apq]$

NOTE The question actually guided you :

$$x = a(p+q) \quad \begin{matrix} \times a \\ x \end{matrix} \quad y = apq \quad \begin{matrix} \times 4a \\ y \end{matrix}$$

$$x^2 = a^2(p+q)^2 \dots (1) \quad 4ay = 4a^2pq \dots (2)$$

$$(1) - (2)$$

$$x^2 - 4ay = a^2(p+q)^2 - 4a^2pq$$

This is also the substitution the question asked for

SOLUTION

$$x^2 - 4ay = [a(p+q)]^2 - 4a(apq)$$

$$= a^2(p+q)^2 - 4a^2pq$$

$$= a^2(p^2 + 2pq + q^2) - 4a^2pq$$

$$= a^2(p^2 + 2pq + q^2 - 4pq)$$

$$= a^2(p^2 - 2pq + q^2)$$

$$= a^2(p-q)^2$$

From (iii) $p-q = 1+pq$

$$= a^2(1+pq)^2$$

$$= a^2\left(1 + \frac{y}{a}\right)^2$$

$$x^2 - 4ay = a^2\left(1 + \frac{2y}{a} + \frac{y^2}{a^2}\right)$$

Now $y = apq$

$$pq = \frac{y}{a}$$

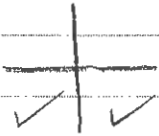
You must do the substitution to gain full marks.

①/2 if correct solution without substitution.

① Note - only 1/2 if they had = 0 and zero if eliminated a^2 using identity (iii)

①/2 subbing in $pq = \frac{y}{a}$

MATHEMATICS EXTENSION I - QUESTION 13

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
$x^2 - 4ay = a^2 + 2ay + y^2$ $x^2 - y^2 = a^2 + 6ay$	$-(\frac{1}{2})$	correct answer
b) $2\sin x - 3\cos x = R\cos(x + \alpha)$ $2\sin x - 3\cos x = R\cos x \cos \alpha - R\sin x \sin \alpha$		
$2 = -R\sin \alpha$ $R\sin \alpha = -2$ $(\frac{1}{2})$ 	$-3 = R\cos \alpha$ $R\cos \alpha = -3$ $(\frac{1}{2})$ 	Many missed negative on $R\sin \alpha = -2$
Overlap quad 3		Many forgot this!
$R^2 \sin^2 \alpha + R^2 \cos^2 \alpha = 4 + 9$ $R^2 (\sin^2 \alpha + \cos^2 \alpha) = 13$ $R^2 = 13$ $R > 0$ $R = \sqrt{13}$	$-(\frac{1}{2})$	
$\frac{R\sin \alpha}{R\cos \alpha} = \frac{-2}{-3}$ $\tan \alpha = \frac{2}{3}$ * QUAD 3 *		
$\alpha = 0.588 \dots$ $\alpha = \pi + 0.588$ $\alpha = 3.73$	$-(1)$	RADIANS! (no penalty but PLEASE remember)
$\sqrt{13} \cos(x + 3.7)$	$-(\frac{1}{2})$	Many forgot to write this

MATHEMATICS EXTENSION I – QUESTION 13

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

$$c) \quad 100 = 80 + Ae^{0.1t}$$

$$t=0$$

$$100 = 80 + Ae^0$$

$$A = 20$$

— (1/2)

Evaluating A

$$N = 80 + 20e^{0.1t}$$

$$200 = 80 + 20e^{0.1t}$$

$$120 = 20e^{0.1t}$$

$$6 = e^{0.1t}$$

$$t = \frac{\ln 6}{0.1}$$

$$t = 17.9175 \dots$$

— (1)

(some halves for very small numerical error)
correct value of t

$t=0$ $t=1$ $t=2$ $t=3$ $t=17$ $t=18$
 2000 2001 2002 2003 ... 2017 2018
 First yr 2nd yr ... 18th yr

During 2017 or in
the 18th year.

(1/2)

Correct year.

MATHEMATICS EXTENSION I – QUESTION 14 TRIAL 2018

pg1. SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(i) Use The sum of the terms of an Arithmetic Sequence to show that $(1+2+3+\dots+n)^2 = \frac{1}{4} n^2 (n+1)^2$ $[1+2+3+\dots+n]$ is an A.S with $a=1$ $d=1$ $n=n$ $L=n$ $S_n = \frac{n}{2} [a+L] \quad \text{OR} \quad S_n = \frac{n}{2} [2a+(n-1)d]$ $= \frac{n}{2} [1+n]$ $= \frac{n}{2} [n+1] \quad \text{--- ①}$ $= \frac{n}{2} [2 + n-1]$ $= \frac{n}{2} [1+n]$ $= \frac{n}{2} [n+1] \quad \text{--- ②}$ substituting ① in $(1+2+3+\dots+n)^2 = \left[\frac{n}{2} (n+1) \right]^2$ $= \frac{n^2}{4} (n+1)^2$ $= \frac{1}{4} (n^2) (n+1)^2$ $\therefore (1+2+3+\dots+n)^2 = \frac{1}{4} (n^2) (n+1)^2$ as required	1	Generally well done (no 1/2 marks)
(ii) Prove by induction $1^3 + 2^3 + \dots + n^3 = (1+2+\dots+n)^2$ for all integers $n \geq 1$ <u>Step 1</u> show the result is true when $n=1$. when $n=1$ LHS = 1^3 RHS = 1^2 $= 1$ $= 1$ LHS = RHS = 1 $\therefore$ the result is true when $n=1$. <u>Step 2</u> Assume the result is true for some integer k (where $k \geq 1$) ie Assume $1^3 + 2^3 + 3^3 + \dots + k^3 = (1+2+3+\dots+k)^2$	1/2	some students used part (i) at this stage justifying and replacing $(1+2+3+\dots+k)^2$ by $\frac{1}{4} k^2 (k+1)^2$

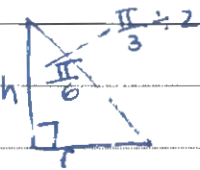
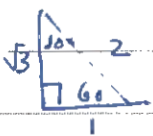
MATHEMATICS EXTENSION I – QUESTION 14 TRIAL 2018

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
pg 2 Step 3 Prove the result is true for $n = k + 1$ assuming it is true for $n = k$. ie Prove $1^3 + 2^3 + 3^3 + \dots + (k+1)^3 = [1 + 2 + 3 + \dots + (k+1)]^2$ LHS = $1^3 + 2^3 + 3^3 + \dots + (k+1)^3$ $= 1^3 + 2^3 + 3^3 + \dots + (k)^3 + (k+1)^3$ $= \underbrace{1^3 + 2^3 + 3^3 + \dots + (k)^3}_{S_k} + (k+1)^3$ $= \frac{1}{4} k^2 (k+1)^2 + (k+1)^3 \quad \left\{ \begin{array}{l} \text{using part (1)} \\ 1 + 2 + 3 + \dots + k^2 = \frac{1}{4} (k^2)(k+1)^2 \end{array} \right.$ $= \frac{(k+1)^2}{4} [k^2 + 4(k+1)]$ $= \frac{(k+1)^2}{4} [k^2 + 4k + 4]$ $= \frac{1}{4} (k+1)^2 (k+2)^2$ $= [1 + 2 + 3 + \dots + (k+1)]^2$ since $\frac{1}{4} (k^2)(k+1)^2 = 1 + 2 + \dots + k^2$ = RHS		some students used part (1) earlier in this stage also.
$\therefore$ the result is true when $n = k$ if it is true when $n = k + 1$. Step 4: if the statement is true for $n = k$, then it is true for $n = k + 1$. since the result is true for $n = 1 + 1 = 2$ and for $n = 2 + 1 = 3$. and so on for all positive integers n.		

MATHEMATICS EXTENSION I – QUESTION 14 TRIAL 2018

pg 3 (b) SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(i) $\begin{array}{r} 2x^2 - 4x - 2 \\ x^2 - 3x + 1 \overline{) 2x^4 - 10x^3 + 12x^2 + 2x - 3} \\ \underline{2x^4 - 6x^3 + 2x^2} \\ -4x^3 + 10x^2 + 2x - 3 \\ \underline{-4x^3 + 12x^2 - 4x} \\ -2x^2 + 6x - 3 \\ \underline{-2x^2 + 6x - 2} \\ -1 \end{array}$	2	-1 per error if had 2 errors but looked basically correct gave $\frac{1}{2}$ instead of 0
(ii) $2x^4 - 10x^3 + 12x^2 + 2x - 3$ $= (x^2 - 3x + 1)(2x^2 - 4x - 2) - 1$ OR $= (x^2 - 3x + 1)[2(x^2 - 2x - 1)] - 1$	1	
(iii) $f(x) = 2x^4 - 10x^3 + 12x^2 + 2x - 3$ $g(x) = x^2 - 3x + 1$ from (ii). $f(x) = g(x)q(x) - 1$ let k be a zero of $g(x)$ then $g(k) = k^2 - 3k + 1$ and $k^2 - 2k + 1 = 0$ ——— ① $\therefore g(k) = 0$ $f(k) = g(k)q(k) - 1 = \text{—————} ②$ { sub ① in = $0 \cdot q(k) - 1$ ② = $0 - 1$ = -1 $\therefore k$ is <u>not</u> a zero of $f(x)$. since $f(k) = -1$. $\therefore f(x)$ and $g(x)$ have no zeros in common.		

MATHEMATICS EXTENSION I – QUESTION 14 TRIAL 2018

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
(i) (i)   $\tan 30^\circ = \frac{1}{\sqrt{3}}$ $\tan \frac{\pi}{6} = \frac{r}{h}$ $\frac{1}{\sqrt{3}} = \frac{r}{h}$ $\therefore r = h \times \frac{1}{\sqrt{3}}$ $r = \frac{h}{\sqrt{3}} \quad \leftarrow \textcircled{1}$	1	most students that had difficulty didn't get the angle, $\frac{\pi}{6}$, correct.
(ii) Show $V = \frac{\pi h^3}{9}$ $V_{\text{cone}} = \frac{1}{3} \pi r^2 h \quad \leftarrow \textcircled{2}$ sub ① in ② $= \frac{1}{3} \times \pi \times \left(\frac{h}{\sqrt{3}}\right)^2 h$ $= \frac{1}{3} \times \pi \times \frac{h^2}{3} \times h$ $\therefore V = \frac{\pi h^3}{9} \quad \text{as required.}$	1.	well done
(iii) $\frac{dV}{dt} = 1.5 \text{ m}^3/\text{s}$ $\frac{dh}{dt} = ?$ $\frac{dV}{dt} = \frac{dV}{dh} \cdot \frac{dh}{dt}$ $V = \frac{\pi h^3}{9}$ $\frac{dV}{dh} = \frac{3\pi h^2}{9}$ $\frac{dV}{dh} = \frac{\pi h^2}{3}$		many methods. 1 or 2 marks. need to show basic understanding and $\frac{dV}{dh}$ to get any marks.

[illegible]

MATHEMATICS EXTENSION I – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
a) LHS = $\frac{\cos x}{1 - \sin x}$ $= \frac{1 - t^2}{1 + t^2}$ $\frac{1 - 2t}{1 + t^2}$ $= \frac{\frac{1 - t^2}{1 + t^2}}{\frac{1 + t^2 - 2t}{1 + t^2}}$ $= \frac{1 - t^2}{(t - 1)^2}$ $= \frac{(1 - t)(1 + t)}{(t - 1)^2}$ $= \frac{-(t - 1)(1 + t)}{(t - 1)^2}$ $= -\frac{(1 + t)}{t - 1}$ $= \frac{-(1 + t)}{-(1 - t)}$ $= \frac{1 + t}{1 - t}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$	
RHS = $\tan\left(45 + \frac{x}{2}\right)$ $= \frac{\tan 45 + \tan \frac{x}{2}}{1 - \tan 45 \tan \frac{x}{2}}$ $= \frac{1 + t}{1 - t} = \text{LHS}$	$\frac{1}{2}$	(2)

MATHEMATICS EXTENSION I - QUESTION 15

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

b) i) Max height occurs when vertical velocity $\dot{y} = 0$.

$$\dot{y} = v \sin \theta - gt$$

$$\therefore v \sin \theta - gt = 0$$

$$t = \frac{v \sin \theta}{g} \quad \text{sub in } y \text{ for Max}$$

Sub in y for Max height

$$y = v \left(\frac{v \sin \theta}{g} \right) \sin \theta - \frac{g}{2} \left(\frac{v \sin \theta}{g} \right)^2$$

$$= \frac{v^2 \sin^2 \theta}{g} - \frac{v^2 \sin^2 \theta}{2g}$$

$$= \frac{2v^2 \sin^2 \theta}{2g} - \frac{v^2 \sin^2 \theta}{2g}$$

$$= \frac{v^2 \sin^2 \theta}{2g} \quad \dots \dots \dots (1)$$

ii) 2) For the second firework the max height reached given $v = \frac{7V}{10}$ and at an angle of 2θ is:

$$y = \frac{\left(\frac{7V}{10} \right)^2 \sin^2 2\theta}{2g} \quad \dots \dots (2)$$

But as both fireworks have the same maximum height: $(1) = (2)$

$$\frac{v^2 \sin^2 \theta}{2g} = \frac{\left(\frac{7V}{10} \right)^2 \sin^2 2\theta}{2g}$$

$$v^2 \sin^2 \theta = \frac{49V^2}{100} \sin^2 2\theta$$

1/2

Some students tried to derive this expression but made a few errors along the way. Care is needed when simplifying.

1 mark to equate the 2 max heights

2

MATHEMATICS EXTENSION I – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
ii) a) Cont'd $100 \sin^2 \theta = 49 \sin^2 2\theta$ $100 \sin^2 \theta = 49 (\sin 2\theta)^2$ $= 49 (2 \sin \theta \cos \theta)^2$ $= 49 \times 4 \times \sin^2 \theta \cos^2 \theta$ $\frac{100}{196} = \cos^2 \theta$ ie, $\cos^2 \theta = \frac{25}{49}$ $\cos \theta = \frac{5}{7} \quad \text{as } \theta \text{ is acute}$	1 $\frac{1}{2}$	(3)
ii) b) The max height is: $y = \frac{v^2 \sin^2 \theta}{2g}$ as $v^2 = 98g$ then $y = \frac{(98g) \times \sin^2 \theta}{2g}$ $= 49 \times \sin^2 \theta$ $= 49 \times (1 - \cos^2 \theta)$ $= 49 \times \left(1 - \frac{25}{49}\right)$ $= 24$	$\frac{1}{2}$ $\frac{1}{2}$	Many student squared 98g and evidently did not receive the correct value for y.
$\therefore$ the maximum height reached by the two fireworks is 24m		(1)

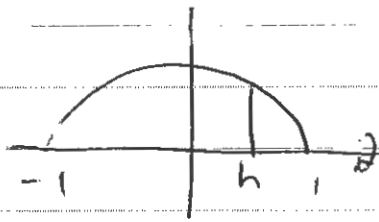
MATHEMATICS EXTENSION I – QUESTION 15

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

c) i)



$$V_1 = \pi \int_{-1}^h (\sqrt{1-x^2})^2 dx$$

$$= \pi \int_{-1}^h (1-x^2) dx$$

$$= \pi \left[x - \frac{x^3}{3} \right]_{-1}^h$$

$$= \pi \left[h - \frac{h^3}{3} - \left(-1 - \frac{(-1)^3}{3} \right) \right]$$

$$= \pi \left[h - \frac{h^3}{3} + 1 - \frac{1}{3} \right]$$

$$= \pi \left(h - \frac{h^3}{3} + \frac{2}{3} \right)$$

$$V_2 = \pi \int_h^1 (1-x^2) dx$$

$$= \pi \left[x - \frac{x^3}{3} \right]_h^1$$

$$= \pi \left[1 - \frac{1}{3} - \left(h - \frac{h^3}{3} \right) \right]$$

$$= \pi \left(\frac{2}{3} - h + \frac{h^3}{3} \right)$$

But $\frac{V_1}{V_2} = \frac{3}{2}$

So $2V_1 = 3V_2$

Remember:

When you are finding the Volume you need to use the formula correctly
i.e. $V = \pi \int y^2 dx$

Many students did not use the π in the formula and lost $\frac{1}{2}$ mark

$\frac{1}{2}$ marks were given for 2 correct volumes V_1 & V_2 .

1 mark for using the ratio

$$\frac{V_1}{V_2} = \frac{3}{2}$$

Correctly.

MATHEMATICS EXTENSION I – QUESTION 15

SUGGESTED SOLUTIONS	MARKS	MARKER'S COMMENTS
c) i) Cont'd $2\left(h - \frac{h}{3} + \frac{2}{3}\right) = 3\left(\frac{2}{3} - h + \frac{h^3}{3}\right)$ $2h - \frac{2h^3}{3} + \frac{4}{3} = 2 - 3h + h^3$ $6h - 2h^3 + 4 = 6 - 9h + 3h^3$ $5h^3 - 15h + 2 = 0$		1/2 mark for correct simplification to establish answer. (3)
Method 2 $V_1 = \pi \int_h^1 (1-x^2) dx$ $= \pi \left(\frac{2}{3} - h + \frac{h^3}{3}\right)$ This volume is $\frac{2}{5}$ of the volume of the sphere. i.e. $V = \frac{2}{5} \times \frac{4}{3} \pi r^3$ $= \frac{8}{15} \pi$		
So $V = V_1$ $\frac{8}{15} \pi = \pi \left(\frac{2}{3} - h + \frac{h^3}{3}\right)$ $8\pi = 10 - 15h + 5h^3$ $5h^3 - 15h + 2 = 0$		

MATHEMATICS EXTENSION I – QUESTION 15

SUGGESTED SOLUTIONS

MARKS

MARKER'S COMMENTS

c.ii) Let $f(x) = 5x^3 - 15x + 2$

$$f'(x) = 15x^2 - 15$$

Taking $h_1 = 0$

$$h_2 = h_1 - \frac{f(h_1)}{f'(h_1)}$$

$$= 0 - \frac{f(0)}{f'(0)}$$

$$= 0 - \frac{2}{-15}$$

$$= \frac{2}{15}$$

} $\frac{1}{2}$

$\frac{1}{2}$

①